t3-w06 — condition C3

Construction — Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees

> C2 plus a validation and repair pass over the draft.

INPUT LAYERS

Input layer	What it carries	Channel	In this condition
Scope	what is to be taught, and to whom	prompt	**present**
Prior knowledge	what the learner already knew coming in	prompt	**present**
Materials	what the lesson was represented with	prompt	**present**
Position	where this objective sat in the sequence	prompt	**present**
Method	the shape of the teaching, as event labels	prompt	**present**
Enacted content	what was actually explained and worked through	prompt	**present**
Set work	what the learners were asked to do in class	prompt	**present**
Assessment	the questions the lesson was checked with	prompt	absent
Figures	the drawings the lesson used	tool	**present**

WHAT THE MODEL IS SENT

Contract `ComposeWriteGroundedSection`, declaring: `artefact_spec, lesson_context, unit_context, lesson_content`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Construction

`subtopic`

Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees

`term`

Third Term

`week`

6

`lesson_objectives`

- construct parallel lines
- construct perpendicular lines
- bisect a given line segment
- construct angle 90°
- construct angle 60°

entry_behaviour

- Students have been taught how to draw lines.

instructional_materials

- Mathematical set.

Objective 1 — construct parallel lines

objective_text

construct parallel lines

current_step

1

instructional_events

• present_content
• elicit_performance

student_tasks

• construct parallel lines

source_figures


```
[
 {
  "caption": "Line $AB$, with a point $P$ marked above it. A set square is drawn below $AB$ with one edge along the line: the right angle is at the left end of that edge, one leg runs down from it, and the hypotenuse closes back to a point on $AB$ to the right.",
  "figure_ref": "fig-01"
 },
 {
  "caption": "The same set square on $AB$, with the ruler now drawn as a long strip lying along the set square's downward edge. The set square's vertical edge and the ruler's right edge are **layered** — only one line is visible on the page. The right angle marked at the corner is the set square's own, between the edge lying along $AB$ and the edge rising from it.",
  "figure_ref": "fig-02"
 },
 {
  "caption": "The same arrangement drawn a second time alongside the first — the set square's edge again **layered** on the ruler's right edge, with no gap between them — and with the ruler leaning — running from upper left to lower right across $AB$ instead of standing vertical. Only $A$ is labelled. The slanted ruler is the point of this panel — it shows that the ruler's inclination does not affect the construction.",
  "figure_ref": "fig-03"
 },
 {
  "caption": "The vertical ruler held fixed, with the set square drawn twice — once with its edge on $AB$ and once higher up, its edge on a second line parallel to $AB$ — and an arrow between the two positions pointing up along the ruler.",
  "figure_ref": "fig-04"
 },
 {
  "caption": "The same slide drawn a second time alongside it, with the leaning ruler. Both set square positions are **layered** on the ruler's right edge, as in fig-03. Here the upper line carries the point $P$, and the set square's upper position has its edge running through $P$.",
  "figure_ref": "fig-05"
 }
]
```


lesson_content

```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "**Examples:** construct a line through Point P so that it is parallel to AB.\n\n(a)
Place a set square so that one edge is accuratly along AB\n\n**fig-01.** Line $AB$, with a point
$P$ marked above it. A set square is drawn below $AB$ with\none edge along the line: the right
angle is at the left end of that edge, one leg runs down from\nit, and the hypotenuse closes back
to a point on $AB$ to the right.\n\n[drawing attached to this section as fig-01, marked A, B, P]
\n\n(b) Place a ruler along one of the other edge of the set square. Use the left hand edge if
you\nare right handed.\n\n**fig-02.** The same set square on $AB$, with the ruler now drawn as a
long strip lying along the\nset square's downward edge. The set square's vertical edge and the
ruler's right edge are\n**layered** — only one line is visible on the page. The right angle
marked at the corner is the\nset square's own, between the edge lying along $AB$ and the edge
rising from it.\n\n[drawing attached to this section as fig-02, marked A, B]\n\n**fig-03.** The
same arrangement drawn a second time alongside the first — the set square's edge\nagain
**layered** on the ruler's right edge, with no gap between them — and with the ruler leaning\n—
running from upper left to lower right across $AB$ instead of standing vertical. Only $A$
is\nlabelled. The slanted ruler is the point of this panel — it shows that the ruler's
inclination\ndoes not affect the construction.\n\n[drawing attached to this section as fig-03,
marked A]\n\n(c) Hold the ruler firmly, slide the set square along the ruler towards P. Stop when
the edge that\nwas on AB reaches P. Draw a line along this edge of the set square through P.
\n\n**fig-04.** The vertical ruler held fixed, with the set square drawn twice — once with its
edge on\n$AB$ and once higher up, its edge on a second line parallel to $AB$ — and an arrow
between the two\npositions pointing up along the ruler.\n\n[drawing attached to this section as
fig-04, marked A, B]\n\n**fig-05.** The same slide drawn a second time alongside it, with the
leaning ruler. Both set\nsquare positions are **layered** on the ruler's right edge, as in
fig-03. Here the upper\nline carries the point $P$, and the set square's upper position has its
edge running through $P$.\n\n[drawing attached to this section as fig-05, marked A, P]"
 }
]
```


Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `assessment_points`, `source_evidence`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

Objective 2 — construct perpendicular lines

objective_text

construct perpendicular lines

current_step

2

instructional_events

• present_content

source_figures


```
[
 {
  "caption": "Line $AB$ with a second line $XY$ crossing it, $X$ above and $Y$ below, and a
right-angle mark in the upper right quadrant at the crossing.",
  "figure_ref": "fig-06"
 },
 {
  "caption": "A plain line with a stroke at each end, labelled $A$ on the left and $B$ on the
right. The point $Q$ named in the instruction is not marked on the drawing.",
  "figure_ref": "fig-07"
 },
 {
  "caption": "Line $AB$ with the set square standing on it as a narrow right-angled triangle: a
short base along $AB$, a near-vertical edge rising from the right end of that base, and the
hypotenuse running up from the left.",
  "figure_ref": "fig-08"
 },
 {
  "caption": "Line $AB$ with $XY$ now drawn across it, $X$ above and $Y$ below, and a right-angle
mark in the upper right quadrant. The foot of $XY$ is not labelled $Q$.",
  "figure_ref": "fig-09"
 }
]
```


lesson_content


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "Two lines are said to be perpendicular to each other when they meet at a right
angle i.e\n$90^\\circ$\n\n**fig-06.** Line $AB$ with a second line $XY$ crossing it, $X$ above
and $Y$ below, and a\nright-angle mark in the upper right quadrant at the crossing.\n\n[drawing
attached to this section as fig-06, marked A, B, X, Y]\n\n1. Draw a line AB mark a point Q on the
line.\n\n**fig-07.** A plain line with a stroke at each end, labelled $A$ on the left and $B$ on
the right.\nThe point $Q$ named in the instruction is not marked on the drawing.\n\n[drawing
attached to this section as fig-07, marked A, B]\n\n2. Place the edge of the set square on line
AB. Let the edge of the ruler meet the hypotenuse of\n   the set square as shown.\n\n**fig-08.**
Line $AB$ with the set square standing on it as a narrow right-angled triangle: a\nshort base
along $AB$, a near-vertical edge rising from the right end of that base, and the\nhypotenuse
running up from the left.\n\n[drawing attached to this section as fig-08, marked A, B]\n\n3.
Pressing firmly on the ruler, slide the set square down along the hypotenuse side until the\n
third edges touches point Q. Draw a line XY\n\n**fig-09.** Line $AB$ with $XY$ now drawn across
```

```
it, $X$ above and $Y$ below, and a right-angle\nmark in the upper right quadrant. The foot of
$XY$ is not labelled $Q$.\n\n[drawing attached to this section as fig-09, marked A, B, X, Y]"
 }
]
```

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `student_tasks`, `activities`, `assessment_points`, `source_evidence`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

Objective 3 — bisect a given line segment

objective_text

bisect a given line segment

current_step

3

instructional_events

• present_content
• elicit_performance

student_tasks

• construct line segment

source_figures


```
[
 {
  "caption": "The line segment $AB$, a stroke at each end, $A$ on the left and $B$ on the
right.",
  "figure_ref": "fig-10"
 },
 {
  "caption": "The segment $AB$ with arc pairs crossing above and below it, the upper crossing
labelled $P$ and the lower one $Q$. Each crossing is drawn as a small X where an arc from $A$
meets an arc from $B$.",
  "figure_ref": "fig-11"
 },
 {
  "caption": "The same arcs with $P$ joined to $Q$, the joining line crossing $AB$ at its middle
and carrying a right-angle mark in the upper right quadrant.",
  "figure_ref": "fig-12"
 }
]
```


lesson_content


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "**Procedures:**\n\n1. Open a pair of compasses so that the radius is about $\
\frac{3}{4}$ of the length of AB.\n2. To bisect the line segment AB means to divide it into two
equal halves\n\n**fig-10.** The line segment $AB$, a stroke at each end, $A$ on the left and $B$
on the right.\n\n[drawing attached to this section as fig-10, marked A, B]\n\n3. Place the sharp
point of the compass on A, Draw two arcs, one above and the other below the\n   midle of AB.
\n\n**fig-11.** The segment $AB$ with arc pairs crossing above and below it, the upper
crossing\nlabelled $P$ and the lower one $Q$. Each crossing is drawn as a small X where an arc
from $A$ meets\nan arc from $B$.\n\n[drawing attached to this section as fig-11, marked A, B, P,
Q]\n\n4. With the same radius, and place the sharp point of the compasses on B Draw two arc so
that they\n   cut the first arc at P and Q\n\n**fig-12.** The same arcs with $P$ joined to $Q$,
the joining line crossing $AB$ at its middle and\ncarrying a right-angle mark in the upper right
quadrant.\n\n[drawing attached to this section as fig-12, marked A, B, P, Q]"
```

```
 }
]
```

Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, activities, assessment_points, source_evidence, objective_learner_state` — carried by the plan or the unit, not sent under this condition.

Objective 4 — construct angle 90°

`objective_text`

construct angle 90°

`current_step`

4

`instructional_events`

• present_content
• elicit_performance

`student_tasks`

• construct angle 90 degrees

`source_figures`

```
[
 {
  "caption": "The line $AB$ with a stroke at each end, the two arc cuts $X$ and $Y$ marked either side of the middle, and the perpendicular $OZ$ erected from $O$ between them, with a right-angle mark at its foot and the two arcs crossing at $Z$ near its top.",
  "figure_ref": "fig-13"
 }
]
```

`lesson_content`

```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "**Procedure:**\n\n1. Draw a line segment $AB$ and mark a point $O$ on it where the right angle is to stand.\n2. With centre $O$ and any convenient radius, draw an arc to cut $AB$ at $X$ and $Y$.\n3. With centre $X$ and a radius greater than $XO$, draw an arc above $AB$.\n4. With centre $Y$ and the same radius as in (3), draw an arc to cut the first arc at $Z$.\n5. Join $OZ$. Then $\\angle ZOA = \\angle ZOB = 90^\\circ$.\n\n**fig-13.** The line $AB$ with a stroke at each end, the two arc cuts $X$ and $Y$ marked either\nside of the middle, and the perpendicular $OZ$ erected from $O$ between them, with a right-angle\nmark at its foot and the two arcs crossing at $Z$ near its top.\n\n[drawing attached to this section as fig-13, marked A, B, X, Y, O, Z]"
 }
]
```

Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, activities, assessment_points, source_evidence, objective_learner_state` — carried by the plan or the unit, not sent under this condition.

Objective 5 — construct angle 60°

objective_text

construct angle 60°

current_step

5

instructional_events

• present_content
• elicit_performance

student_tasks

• construct angle 60 degrees

source_figures


```
[
 {
  "caption": "The line $BC$ with $B$ at the left, a stroke marking $X$ part way along, a long arc swung from $B$ through $X$ and up to the left, a shorter arc swung from $X$ cutting it, and a straight line from $B$ to that crossing. The crossing itself is not labelled.",
  "figure_ref": "fig-14"
 }
]
```


lesson_content


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "**Procedure:**\n\n1. Draw a line segment BC with centre B at any convinent radius. Draw an arc to cut BC at X\n2. with centre X and the same radius as in (i) draw an arc to cut the first arc at A\n3. Join AB.\n\n**fig-14.** The line $BC$ with $B$ at the left, a stroke marking $X$ part way along, a long arc\nswung from $B$ through $X$ and up to the left, a shorter arc swung from $X$ cutting it, and a\nstraight line from $B$ to that crossing. The crossing itself is not labelled.\n\n[drawing attached to this section as fig-14, marked B, C, X]"
 }
]
```


Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `assessment_points`, `source_evidence`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

WHAT THE MODEL PRODUCED

Constructing a Parallel Line through a Point

EXPLANATION

You already know how to draw straight lines. To draw a line through a given point \(P\) that is parallel to an existing line \(AB\), use a set-square and a ruler:

1. Place the set-square so that one of its edges lies exactly on the line \(AB\) (see fig-01). The right-angle corner of the set-square should be at the left end of that edge.
2. Take a ruler and line it up with another edge of the set-square – the edge that is perpendicular to the one on \(AB\) (fig-02). Hold the ruler firmly.
3. Keeping the ruler fixed, slide the set-square along the ruler until the edge that was on \(AB\) reaches the point \(P\). The set-square does not turn; it only translates (fig-04 and fig-05 illustrate this slide).
4. Draw a line along the edge of the set-square that now passes through \(P\). This new line is parallel to \(AB\) because the set-square kept the same orientation throughout the slide.

The position of the ruler does not matter – whether it is horizontal or leaning (fig-03) – as long as it stays still while you slide the set-square.

Remember: the key idea is that sliding a set-square along a fixed straight edge moves the whole shape without rotating it, so the edge that was on \(AB\) stays parallel to \(AB\) when it reaches \(P\).

WORKED_EXAMPLE

Worked example

Given: line \(AB\) is drawn horizontally and point \(P\) is above the line as shown in fig-01.

Step 1: Place the set-square so that its lower edge lies on \(AB\) (the edge shown in fig-01).

Step 2: Align a ruler with the short upright edge of the set-square (the edge that makes the right angle with \(AB\)). Hold the ruler steady.

Step 3: While keeping the ruler fixed, slide the set-square straight up along the ruler until the lower edge reaches point \(P\). The set-square moves without turning – this is the slide shown in fig-04 and fig-05.

Step 4: Draw a line along the lower edge of the set-square now passing through \(P\). The line you draw is \(PQ\).

Result: \(PQ\) is parallel to \(AB\). You can check this by noting that the two lines never meet and that the corresponding angles formed with the ruler are equal (both are right angles).

VISUAL

```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (7,0);
  \node[left] at (0,0) {$A$};
  \node[above] at (7,0) {$B$};
  \fill (3.4,2.8) circle (1.6pt);
  \node[above] at (3.4,2.8) {$P$};
  \draw (2.7,0) -- (2.7,-1.1) -- (5.2,0);
  \draw (2.7,-0.26) -- (2.96,-0.26) -- (2.96,0);
\end{tikzpicture}
```

VISUAL

```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (8,0);
  \node[below left] at (0,0) {$A$};
  \node[above] at (8,0) {$B$};
  \draw[rounded corners=3pt] (1.29,3.50) rectangle (1.72,-4.12);
  % The set square's vertical edge lies ALONG the ruler's right edge (x=1.72) — in the
  % source the two are layered, so only one line is visible. Drawn collinear, not offset.
  \draw (1.72,0) -- (1.72,-1.9) -- (5.4,0);
```

```
  \draw (1.72,-0.26) -- (1.98,-0.26) -- (1.98,0);
\end{tikzpicture}
```

```
VISUAL
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (8,0);
  \node[below left] at (0,0) {$A$};
  \node[above right] at (8,0) {$B$};
  % The ruler now lies along the sloping edge, extending past it at both ends. Drawn a
  % hairline clear of that edge so the set square stays visible as a triangle.
  \draw[rounded corners=2pt] (-0.171,-3.033) -- (6.760,0.545) -- (6.967,0.145) -- (0.036,-3.433) -- cycle;
  \draw (1.72,0) -- (1.72,-1.9) -- (5.4,0);
  \draw (1.72,-0.26) -- (1.98,-0.26) -- (1.98,0);
\end{tikzpicture}
```

```
VISUAL
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (8,0);
  \node[below left] at (0,0) {$A$};
  \node[above] at (8,0) {$B$};
  \draw (0.71,2.40) -- (8,2.40);
  \draw[rounded corners=3pt] (0.28,3.65) rectangle (0.71,-3.97);
  % ONE set square drawn in two positions, translated straight up the ruler's right edge
  % (x=0.71). Both are congruent — leg 1.21, width 2.68 — so the two hypotenuses are
  % exactly parallel, which is the visible signature of a slide.
  \draw (0.71,0) -- (0.71,-1.21) -- (3.39,0);
  \draw (0.71,-0.26) -- (0.97,-0.26) -- (0.97,0);
  \draw (0.71,2.40) -- (0.71,1.19) -- (3.39,2.40);
  \draw (0.71,2.14) -- (0.97,2.14) -- (0.97,2.40);
  \draw[->] (2.44,0.1) -- (2.44,1.37);
\end{tikzpicture}
```

```
VISUAL
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (8,0);
  \node[below left] at (0,0) {$A$};
  \node[above right] at (8,0) {$B$};
  \draw (0.35,2.2) -- (8,2.2);
  \fill (6.0,2.2) circle (1.6pt);
  \node[above] at (6.0,2.2) {$P$};
  % The ruler is held still; it spans both positions of the set square, drawn a hairline
  % clear of their sloping edges.
  \draw[rounded corners=2pt] (-0.026,-2.119) -- (7.616,2.247) -- (7.824,1.882) -- (0.182,-2.484) -- cycle;
  \draw (0.6,0) -- (0.6,-1.6) -- (3.4,0);
  \draw (0.6,-0.26) -- (0.86,-0.26) -- (0.86,0);
  % The same set square, translated along the ruler until its edge reaches P.
  \draw (4.45,2.2) -- (4.45,0.6) -- (7.25,2.2);
  \draw (4.45,1.94) -- (4.71,1.94) -- (4.71,2.2);
  \draw[->] (2.6,0.55) -- (4.0,1.35);
\end{tikzpicture}
```

1. Using a set-square and a ruler, construct a line through point \(C\) that is parallel to the given line \(DE\). Sketch your construction steps on paper.

answer: A line through \(C\) that never meets \(DE\) and is drawn by sliding the set-square as described.

2. A line \(XY\) is drawn on your paper. Point \(Z\) lies to the right of the line. Construct a line through \(Z\) that is parallel to \(XY\). Write down the steps you would follow.

answer: A line through \(Z\) that stays the same direction as \(XY\).

How to Construct a Perpendicular Line

EXPLANATION

A line is perpendicular to another when they meet at a right angle (\(90^\circ\)). To draw a line that is perpendicular to a given line \(AB\) at a point \(Q\) on \(AB\), you can use a set square:

1. Draw the base line \(AB\) and mark the point \(Q\) where the perpendicular will pass.
2. Place the edge of the set square along \(AB\) so that the short base of the set square lies on the line. The upright edge of the set square will be at a right angle to \(AB\).
3. Slide the set square down the hypotenuse side until the third edge (the one opposite the right-angle corner) touches the point \(Q\).
4. Draw a line through the two points where the upright edge meets the paper – this is the line \(XY\) and it is perpendicular to \(AB\) at \(Q\).

Figure references:
- See **fig-06** for the definition of a right-angle.
- **fig-07** shows the base line \(AB\).
- **fig-08** illustrates the set square positioned on \(AB\).
- **fig-09** shows the completed perpendicular line \(XY\).

WORKED_EXAMPLE

Worked example

Suppose you have a line \(AB\) that is 8 cm long and you need a perpendicular line through a point \(Q\) that is 3 cm from \(A\).

1. Draw \(AB\) and label the ends \(A\) and \(B\). Mark point \(Q\) on the line, 3 cm from \(A\).
2. Place the set square so that its short base lies on \(AB\) and the right-angle corner is at the right end of the base.
3. Slide the set square down the hypotenuse until the third edge touches \(Q\).
4. Draw the line through the two points where the upright edge meets the paper. This new line \(XY\) is perpendicular to \(AB\) at \(Q\).

You now have a right-angle at \(Q\) as shown in **fig-09**.

VISUAL

```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (-0.5,0) -- (7.5,0);
  \draw (0,0.25) -- (0,-0.25);
  \draw (5,0.25) -- (5,-0.25);
  \node[below] at (0,-0.3) {$A$};
  \node[below] at (5,-0.3) {$B$};
  \draw (2.5,2.6) -- (2.5,-2.6);
  \node[above] at (2.5,2.6) {$X$};
  \node[below] at (2.5,-2.6) {$Y$};
  \draw (2.82,0) -- (2.82,0.32) -- (2.5,0.32);
\end{tikzpicture}
```

VISUAL

```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (7,0);
  \draw (0,0.22) -- (0,-0.22);
  \draw (7,0.22) -- (7,-0.22);
  \node[below] at (0,-0.28) {$A$};
  \node[below] at (7,-0.28) {$B$};
\end{tikzpicture}
```

VISUAL

```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (8,0);
  \node[below] at (0.41,-0.05) {$A$};
  \node[above] at (3.93,0.05) {$B$};
  % The upright edge stands square on the base: step 2 and 3 name a hypotenuse and a
```

```
  % "third edge", which only a marked right angle lets the reader tell apart.
  \draw (1.54,0) -- (2.25,1.0) -- (2.25,0) -- cycle;
  \draw (1.97,0) -- (1.97,0.28) -- (2.25,0.28);
\end{tikzpicture}
```

```
VISUAL
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (8,0);
  \node[below] at (0.15,-0.1) {$A$};
  \node[above] at (4.52,0.1) {$B$};
  \draw (2.1,1.85) -- (2.15,-0.8);
  \node[above] at (2.1,1.85) {$X$};
  \node[below] at (2.15,-0.8) {$Y$};
  \draw (2.44,0) -- (2.44,0.32) -- (2.12,0.32);
\end{tikzpicture}
```

1. 1. Draw a line \(AB\) 10 cm long. Mark a point \(Q\) 4 cm from \(A\). Using a set square, construct a line through \(Q\) that is perpendicular to \(AB\).

answer: A perpendicular line \(XY\) drawn through \(Q\) meeting \(AB\) at a right angle.

2. 2. On a piece of paper, draw a horizontal line and label it \(AB\). Choose any point \(Q\) on the line. Show the steps to draw a line perpendicular to \(AB\) at \(Q\) using a set square.

answer: Place the set square on \(AB\), slide it until the third edge touches \(Q\), then draw the upright edge as the perpendicular line.

3. 3. A line \(AB\) is drawn, and a point \(Q\) is marked 6 cm from \(A\). After constructing the perpendicular line through \(Q\), what is the measure of the angle between \(AB\) and the new line?

answer: 90°

How to Bisect a Line Segment

EXPLANATION

To bisect a line segment you need to find its exact midpoint so that the two new parts are equal in length. You will use a compass and a ruler. First, open the compass to a width a little larger than half the length of the segment (about three-quarters of the segment's length). Place the compass point on one end of the segment and draw an arc above and another below the line. Then, without changing the compass width, place the point on the other end and draw two more arcs that intersect the first pair. The two intersection points are labelled \(P\) (above) and \(Q\) (below). Finally, join \(P\) and \(Q\) with a straight line. This line cuts the original segment at its midpoint \(M\) and does so at a right angle. See figures fig-10, fig-11 and fig-12 for a visual guide.

VISUAL

```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (6,0);
  \draw (0,0.22) -- (0,-0.22);
  \draw (6,0.22) -- (6,-0.22);
  \node[below left] at (0,-0.1) {$A$};
  \node[below right] at (6,-0.1) {$B$};
  \draw (0,0) ++(37:4.5) arc (37:59:4.5);
  \draw (6,0) ++(121:4.5) arc (121:143:4.5);
  \draw (0,0) ++(-59:4.5) arc (-59:-37:4.5);
  \draw (6,0) ++(-143:4.5) arc (-143:-121:4.5);
  \draw (3,3.35) -- (3,-3.35);
  \node[above] at (3,3.35) {$P$};
  \node[below] at (3,-3.35) {$Q$};
  \node[below right] at (3,-0.1) {$M$};
  \draw (3.32,0) -- (3.32,0.32) -- (3,0.32);
\end{tikzpicture}
```

VISUAL

```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (6,0);
  \draw (0,0.22) -- (0,-0.22);
  \draw (6,0.22) -- (6,-0.22);
  \node[below left] at (0,-0.1) {$A$};
  \node[below right] at (6,-0.1) {$B$};
  \draw (0,0) ++(37:4.5) arc (37:59:4.5);
  \draw (6,0) ++(121:4.5) arc (121:143:4.5);
  \draw (0,0) ++(-59:4.5) arc (-59:-37:4.5);
  \draw (6,0) ++(-143:4.5) arc (-143:-121:4.5);
  \node[above] at (3,3.5) {$P$};
  \node[below] at (3,-3.5) {$Q$};
\end{tikzpicture}
```

VISUAL

```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (7,0);
  \draw (0,0.22) -- (0,-0.22);
  \draw (7,0.22) -- (7,-0.22);
  \node[below left] at (0,-0.1) {$A$};
  \node[below right] at (7,-0.1) {$B$};
\end{tikzpicture}
```

WORKED_EXAMPLE

Worked example (using the diagram in fig-10 to fig-12):

1. **Set the compass** – Open it to about \(\frac{3}{4}\) of the length of \(AB\).
2. **First arcs** – Place the compass point on \(A\) and draw an arc above the line and another below it.
3. **Second arcs** – Without changing the radius, place the point on \(B\) and draw two arcs that intersect the first pair. The intersections are \(P\) (above) and \(Q\) (below) as shown in fig-11.
4. **Join the intersections** – Use a ruler to draw the line \(PQ\). This is the perpendicular bisector of \(AB\).
5. **Locate the midpoint** – The line \(PQ\) meets \(AB\) at \(M\). Because \(PQ\) is a perpendicular bisector, \(M\) is exactly halfway

> between \(A\) and \(B\); therefore \(AM = MB\). Figure fig-12 shows the completed construction with the right-angle mark at \(M\).
>
> You have now bisected the segment \(AB\).

1. Given a line segment \(CD\) that is 8 cm long, describe the steps you would take to bisect it using a compass and ruler. (You do not need to draw the diagram, just list the steps.)

answer: 1. Open the compass to a width a little larger than 4 cm (about 6 cm). 2. Place the compass point on \(C\) and draw an arc above and below \(CD\). 3. Without changing the radius, place the point on \(D\) and draw two arcs that intersect the first pair; label the intersections \(P\) and \(Q\). 4. Draw the line \(PQ\) with a ruler. 5. The line \(PQ\) cuts \(CD\) at its midpoint \(M\), giving \(CM = MD = 4\) cm.

2. You have drawn a segment \(EF\). After constructing the perpendicular bisector, you find that the point where it meets \(EF\) is not exactly halfway. What could be a possible reason, and how would you correct it?

answer: A possible reason is that the compass width was changed between drawing the arcs from \(E\) and \(F\), so the arcs did not intersect symmetrically. To correct it, keep the compass opening the same for both ends, redraw the arcs, and then join the new intersection points to get the correct bisector.

Constructing a Right Angle (90°)

EXPLANATION

To draw a right angle at a point on a line you need only a ruler and a pair of compasses.

Steps

1. Draw a straight line \(AB\) and pick the point \(C\) on it where the right angle is required.
2. Place the compass point at \(C\) and draw an arc that cuts the line at two points, label them \(D\) and \(E\).
3. Without changing the compass width, move the point to \(D\) and draw a larger arc above the line.
4. Keeping the same width, place the compass at \(E\) and draw another arc that meets the previous one at \(F\).
5. Join \(C\) to \(F\) with a ruler.
6. The line \(CF\) is perpendicular to \(AB\); therefore \(\angle FCB = 90^{\circ}\).

The construction is illustrated in **fig-13**.

WORKED_EXAMPLE

Worked example

Suppose you need a right angle at point \(C\) on the line \(AB\).

- Draw \(AB\) and mark \(C\).
- With the compasses set to any convenient radius, draw an arc centred at \(C\) intersecting \(AB\) at \(D\) and \(E\).
- Increase the compass width a little (larger than \(CD\)).
- With the point at \(D\), draw an arc above the line.
- With the same width, draw an arc from \(E\); the arcs cross at \(F\).
- Connect \(C\) to \(F\). The line \(CF\) is now perpendicular to \(AB\), giving a perfect \(90^{\circ}\) angle at \(C\).

Check your work by seeing that the right-angle mark (a small square) appears at \(C\) in the diagram.

VISUAL

```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (7,0);
  \draw (0,0.22) -- (0,-0.22);
  \draw (7,0.22) -- (7,-0.22);
  \node[below left] at (0,-0.1) {$A$};
  \node[below right] at (7,-0.1) {$B$};
  \draw (0.29,0.22) -- (0.29,-0.22);
  \draw (0.84,0.34) -- (0.84,-0.34);
  \draw (6.13,0.22) -- (6.13,-0.22);
  \draw (6.60,0.34) -- (6.60,-0.34);
  \draw (3.8,0) -- (3.8,3.2);
  \draw (4.12,0) -- (4.12,0.32) -- (3.8,0.32);
  \draw (2.9,2.55) -- (4.7,3.05);
  \draw (2.9,3.05) -- (4.7,2.55);
\end{tikzpicture}
```

1. Construct a right angle at point \(C\) on the line \(AB\) using only a ruler and compasses. List the steps you would follow.

answer: 1. Draw line AB and mark point C. 2. With compasses at C draw an arc cutting AB at D and E. 3. Widen compasses, draw arc from D above the line. 4. With same width draw arc from E intersecting previous arc at F. 5. Join C to F; CF is perpendicular to AB, giving a 90° angle at C.

2. You have drawn line XY and need a right angle at point Z on the line. After drawing the first arc that meets the line at points P and Q, you accidentally set the compass to the same width as the first arc and draw the second arcs from P and Q. Will the construction give a right angle? Explain why or why not, and state the correct adjustment.

answer: No. Using the same width as the first arc makes the two new arcs meet only at Z, so no point F is created to define the perpendicular. The compass must be opened wider than the first radius before drawing the arcs from P and Q.

Objective 5

Constructing a 60° Angle with a Compass and Straightedge

EXPLANATION

To draw a 60° angle you only need a straightedge and a compass.

1. **Draw a base line** – Mark a point B and draw a horizontal line BC of any convenient length.
2. **Mark a point on the line** – Choose a point X on BC (see figure \[fig-14\]) and place the compass point at B. Open the compass to any radius and draw an arc that cuts BC at X.
3. **Transfer the radius** – Without changing the compass width, place the compass point at X and draw a second arc that intersects the first arc. Call the intersection A.
4. **Join the points** – Draw the straight line AB. The angle $\angle ABX$ is exactly 60°.

The construction works because the two arcs create an equilateral triangle ABX, and each interior angle of an equilateral triangle measures 60°.

Figure \[fig-14\] shows the line BC, the point X, the two arcs and the resulting line AB.

VISUAL

```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (7,0);
  \draw (0,0.22) -- (0,-0.22);
  \node[below left] at (0,-0.05) {$B$};
  \node[above right] at (7,0) {$C$};
  \draw (5.17,0.22) -- (5.17,-0.22);
  \node[below] at (5.17,-0.25) {$X$};
  \draw (0,0) ++(0:5.17) arc (0:74:5.17);
  \draw (5.17,0) ++(103:5.17) arc (103:140:5.17);
  \draw (0,0) -- (2.59,4.48);
\end{tikzpicture}
```

WORKED_EXAMPLE

Worked example

Suppose you need a 60° angle at point B on a line segment BC that is 8 cm long.

- Draw BC and mark B at the left end.
- Set the compass to 4 cm (half the length of BC) and draw an arc from B intersecting BC at X (so $BX = 4$ cm).
- Keeping the compass at 4 cm, place the point at X and draw a second arc that meets the first arc at A.
- Connect A to B with a straight line.

The angle $\angle ABX$ you have just drawn is 60°.

You can verify the construction by measuring the sides AB, BX, and AX – they are all equal, confirming the equilateral triangle and the 60° angle.

1. Using a ruler and compass, construct a 60° angle on a line segment DE that is 10 cm long. Describe the steps you would follow.

answer: 1. Draw line DE and mark D at the left end. 2. Set the compass to any convenient radius (e.g., 5 cm) and draw an arc from D cutting DE at X. 3. Without changing the radius, place the compass point at X and draw a second arc intersecting the first arc at A. 4. Draw the line AD. The angle $\angle ADX$ is 60°.

2. You have drawn a line MN and marked a point P on it. With a compass set to the distance MP, you draw an arc from M and another arc from P that intersect at Q. What is the measure of $\angle QMP$? Explain why.

answer: $\angle QMP = 60^\circ$ because the arcs create an equilateral triangle MQP, and each interior angle of an equilateral triangle is 60°.

LAYER BY LAYER: WHAT WAS GROUNDED AND WHAT WAS NOT

Output layer	Made	Written from	Verdict

Title the reader-facing framing of the section	5	Scope	**grounded**
Explanation the principle, rule or relationship to review	5	Enacted content	**grounded**
Worked example a problem with its solution worked through	5	Enacted content	**grounded**
Figure a drawing the learner reads from	14	Figures	**grounded**
Practice what the learner does	11	Set work	**grounded**
Answer what the learner checks against	11	Enacted content	**grounded**